

Q1

(i)

$$\frac{dx}{dt} = k(P - x), k > 0$$

$$\int \frac{1}{P - x} dx = \int k dt$$

$$-\ln(P - x) = kt + C \quad \text{since } P > x$$

$$\ln(P - x) = -kt - C$$

$$P - x = e^{-kt - C}$$

$$P - x = e^{-C} \cdot e^{-kt}$$

$$P - x = Ae^{-kt}, \quad A = e^{-C}$$

$$x = P - Ae^{-kt}$$

$$\text{When } t = 0, \quad x = 0 \Rightarrow A = P$$

$$x = P - Pe^{-kt}$$

$$x = P(1 - e^{-kt}) \quad (\text{Shown})$$

(ii)

$$\text{When } t = 12, \quad x = \frac{1}{2}P$$

$$\frac{1}{2}P = P(1 - e^{-12k})$$

$$e^{-12k} = \frac{1}{2}$$

$$-12k = \ln \frac{1}{2} = -\ln 2$$

$$k = \frac{1}{12} \ln 2$$

$$x = P \left(1 - e^{-\frac{t}{12} \ln 2} \right)$$

$$\text{When } x = 0.8P,$$

$$0.8P = P \left(1 - e^{-\frac{t}{12} \ln 2} \right)$$

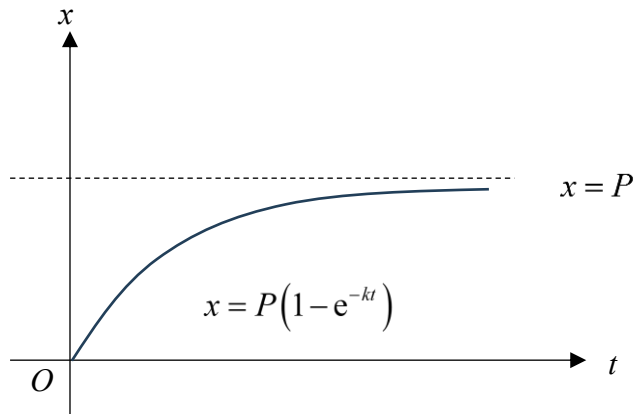
$$e^{-\frac{t}{12} \ln 2} = 0.2$$

$$-\frac{t}{12} \ln 2 = \ln 0.2$$

$$t = \frac{-12 \ln 0.2}{\ln 2}$$

$$t = 27.9 \approx 28 \text{ hours (nearest hour)}$$

(iii)

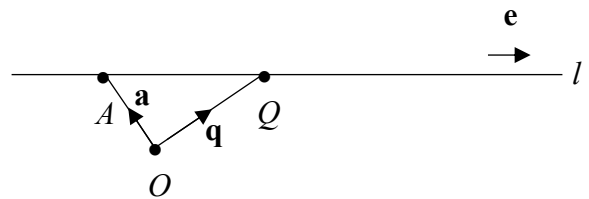


Q2

(i)

Method 1:Equation of l : $\mathbf{r} = \mathbf{a} + \lambda \mathbf{e}$, $\lambda \in \mathbb{R}$ Since Q lies on l , $\mathbf{q} = \mathbf{a} + \lambda \mathbf{e}$, for some value of $\lambda \in \mathbb{R}$

$$\begin{aligned}
 (\mathbf{q} - \mathbf{a}) \times \mathbf{e} &= (\mathbf{a} + \lambda \mathbf{e} - \mathbf{a}) \times \mathbf{e} \\
 &= \lambda \mathbf{e} \times \mathbf{e} \\
 &= \mathbf{0} \text{ (Shown)}
 \end{aligned}$$

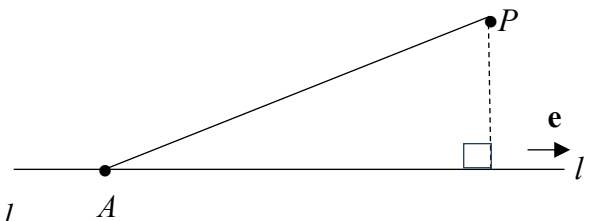
**Method 2:**

$$\overrightarrow{AQ} = \mathbf{q} - \mathbf{a}$$

Since, $\overrightarrow{AQ} \parallel \mathbf{e}$, $(\mathbf{q} - \mathbf{a}) \times \mathbf{e} = \mathbf{0}$. (Shown)

(ii)

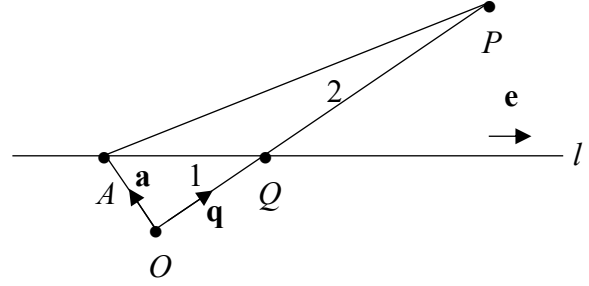
$$|(\mathbf{p} - \mathbf{a}) \times \mathbf{e}| = |\overrightarrow{AP} \times \mathbf{e}|$$

 $|(\mathbf{p} - \mathbf{a}) \times \mathbf{e}|$ represent the perpendicular distance from P to l .**OR** area of parallelogram with adjacent sides parallel and equal in magnitude to $\overrightarrow{AP}$ and $\mathbf{e}$

(iii)

Method 1:

$$\begin{aligned}
 \text{Area of triangle } APQ &= \frac{1}{2} |(\mathbf{p} - \mathbf{a}) \times \mathbf{e}| |\overrightarrow{AQ}| \\
 &= \frac{1}{2} |(3\mathbf{q} - \mathbf{a}) \times \mathbf{e}| (2) \\
 &= |(2\mathbf{q} + \mathbf{q} - \mathbf{a}) \times \mathbf{e}| \\
 &= |(2\mathbf{q}) \times \mathbf{e} + (\mathbf{q} - \mathbf{a}) \times \mathbf{e}| \\
 &= |2(\mathbf{q} \times \mathbf{e}) + \mathbf{0}| \quad (\text{Using (i)}) \\
 &= 2|\mathbf{q} \times \mathbf{e}|
 \end{aligned}$$

**Method 2:**

$$\begin{aligned}
 \text{Area of triangle } APQ &= \frac{1}{2} |(\mathbf{p} - \mathbf{a}) \times \mathbf{e}| |\overrightarrow{AQ}| \\
 &= \frac{1}{2} |(3\mathbf{q} - \mathbf{a}) \times \mathbf{e}| (2) \\
 &= |(3\mathbf{q} - \mathbf{q} + \lambda\mathbf{e}) \times \mathbf{e}| \quad (\text{Since } \mathbf{q} = \mathbf{a} + \lambda\mathbf{e} \Rightarrow \mathbf{a} = \mathbf{q} - \lambda\mathbf{e}) \\
 &= |(2\mathbf{q}) \times \mathbf{e} + \lambda\mathbf{e} \times \mathbf{e}| \\
 &= |2(\mathbf{q} \times \mathbf{e}) + \mathbf{0}| \\
 &= 2|\mathbf{q} \times \mathbf{e}|
 \end{aligned}$$

Method 3:

$$\begin{aligned}
 \text{Area of triangle } APQ &= \frac{1}{2} |\overrightarrow{AQ} \times \overrightarrow{QP}| \quad \text{or} \quad \frac{1}{2} |\overrightarrow{AQ} \times \overrightarrow{AP}| \quad \text{or} \quad \frac{1}{2} |\overrightarrow{AP} \times \overrightarrow{PQ}| \\
 &= \frac{1}{2} |(\mathbf{q} - \mathbf{a}) \times (\mathbf{p} - \mathbf{q})| \\
 &= \frac{1}{2} |(\lambda\mathbf{e}) \times (3\mathbf{q} - \mathbf{q})| \quad (\text{Since } \mathbf{q} = \mathbf{a} + \lambda\mathbf{e} \Rightarrow \mathbf{q} - \mathbf{a} = \lambda\mathbf{e}) \\
 &= \frac{1}{2} |\lambda| |\mathbf{e} \times (2\mathbf{q})| \\
 &= 2|\mathbf{q} \times \mathbf{e}| \quad (\text{Since } |\overrightarrow{AQ}| = 2, |\lambda\mathbf{e}| = 2 \Rightarrow |\lambda| = 2)
 \end{aligned}$$

Note: $\overrightarrow{AQ} = \mathbf{q} - \mathbf{a} = \lambda\mathbf{e}$
 Since $|\overrightarrow{AQ}| = 2$,
 $\overrightarrow{AQ} = 2\mathbf{e}$ or $-2\mathbf{e}$
 Hence, $|\lambda| = 2$

(iv)

Method 1:

$$\text{Area} = 2|\mathbf{q} \times \mathbf{e}| = 3$$

$$2|\mathbf{q}||\mathbf{e}|\sin\theta = 3$$

$$2(2)(1)\sin\theta = 3$$

$$\sin\theta = \frac{3}{4}$$

$$\theta = 48.590^\circ \approx 48.6^\circ \quad (1\text{dp})$$

Acute angle between l and PQ = Acute angle between $\mathbf{q}$ and $\mathbf{e} = 48.6^\circ$

Method 2:

$$\text{Area} = \frac{1}{2}|\overrightarrow{AQ}||\overrightarrow{QP}|\sin\theta = 3$$

$$\frac{1}{2}(2)|\mathbf{p} - \mathbf{q}|\sin\theta = 3$$

$$|2\mathbf{q}|\sin\theta = 3 \quad \text{Since } \mathbf{p} = 3\mathbf{q}$$

$$2|\mathbf{q}|\sin\theta = 3$$

$$\sin\theta = \frac{3}{2(2)}$$

$$\theta = 48.590^\circ \approx 48.6^\circ$$

Acute angle between l and PQ = Acute angle between $\mathbf{q}$ and $\mathbf{e} = 48.6^\circ$

Q3

(i)

$$x = \sin\theta + 1 \quad \text{and} \quad y = \sqrt{3}\cos\theta - 1$$

$$\frac{dx}{d\theta} = \cos\theta \quad \text{and} \quad \frac{dy}{d\theta} = -\sqrt{3}\sin\theta$$

$$\frac{dy}{dx} = \frac{-\sqrt{3}\sin\theta}{\cos\theta} = -\sqrt{3}\tan\theta \quad (\text{Shown})$$

(ii)

$$\text{At } T(\sin t + 1, \sqrt{3}\cos t - 1), \quad \theta = t$$

$$\text{Tangent at } T \text{ makes an angle of } \frac{3\pi}{4} \text{ with the positive } x\text{-axis: } \frac{dy}{dx} = -\sqrt{3}\tan t = \tan \frac{3\pi}{4} = -1$$

$$\tan t = \frac{1}{\sqrt{3}}$$

$$t = \frac{\pi}{6}$$

$$\text{Hence, } T\left(\sin \frac{\pi}{6} + 1, \sqrt{3} \cos \frac{\pi}{6} - 1\right) = \left(\frac{3}{2}, \frac{1}{2}\right)$$

Gradient of normal at $T = 1$

$$\text{Equation of normal at } T: y - \frac{1}{2} = (1)\left(x - \frac{3}{2}\right)$$

$$y = x - 1$$

(iii)

For tangent to C to be parallel to the x -axis, $\frac{dy}{dx} = -\sqrt{3} \tan \theta = 0 \Rightarrow \theta = 0$

$$\text{Hence, } y = \sqrt{3} \cos 0 - 1 = \sqrt{3} - 1$$

(iv)

At Q ,

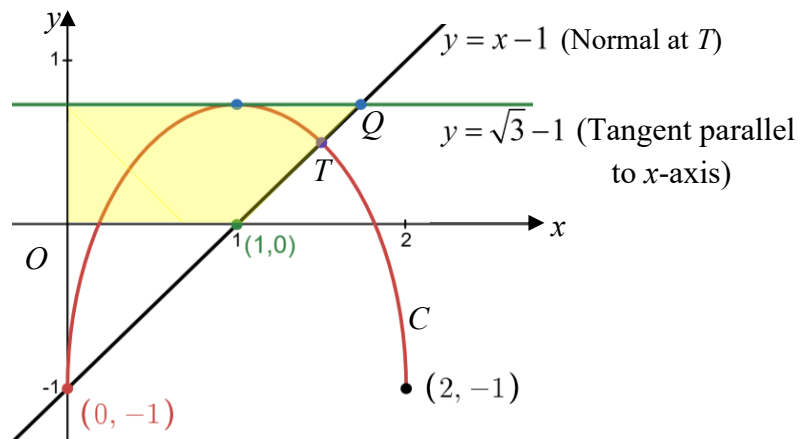
$$x - 1 = \sqrt{3} - 1$$

$$x = \sqrt{3}$$

Area of quadrilateral

$$= \frac{1}{2}(\sqrt{3} + 1)(\sqrt{3} - 1)$$

$$= 1$$



Q4

(a)

Method 1

Since polynomial has only real coefficients, if $3 - i$ is a root, $3 + i$ is another root.

$$[w - (3 - i)][w - (3 + i)]$$

$$= [(w - 3) + i][(w - 3) - i]$$

$$= (w - 3)^2 - i^2$$

$$= w^2 - 6w + 10$$

$$5w^3 + pw^2 + 68w + q = (w^2 - 6w + 10)(5w + a)$$

$$= 5w^3 + aw^2 - 30w^2 - 6aw + 50w + 10a$$

Comparing coefficients of

$$w^2: a - 30 = p \text{ --- (1)}$$

$$w: -6a + 50 = 68 \text{ --- (2)}$$

$$\text{Constant: } 10a = q \text{ --- (3)}$$

$$\text{From (2): } -6a + 50 = 68 \Rightarrow a = -3$$

$$\text{Subt into (1): } -3 - 30 = p \Rightarrow p = -33$$

$$\text{Subt into (3): } q = -30$$

$$5w^3 - 33w^2 + 68w - 30 = (w^2 - 6w + 10)(5w - 3) = 0$$

The other roots are $3 + i$ and $\frac{3}{5}$.

Method 2

$$w = 3 - i$$

$$w^2 = (3 - i)^2 = 9 - 6i + i^2 = 8 - 6i$$

$$w^3 = (3 - i)(8 - 6i) = 24 - 18i - 8i + 6i^2 = 18 - 26i$$

$$5w^3 + pw^2 + 68w + q = 0$$

$$5(18 - 26i) + p(8 - 6i) + 68(3 - i) + q = 0$$

$$90 - 130i + 8p - 6pi + 204 - 68i + q = 0$$

Comparing real and imaginary parts:

$$\text{Real: } 90 + 8p + 204 + q = 0 \text{ --- (1)}$$

$$\text{Im: } -130 - 6p - 68 = 0 \Rightarrow p = -33$$

Sub into (1)

$$90 + 8(-33) + 204 + q = 0 \Rightarrow q = -30$$

Since polynomial has only real coefficients, if $3 - i$ is a root, $3 + i$ is another root.

$$[w - (3 - i)][w - (3 + i)]$$

$$= [(w - 3) + i][(w - 3) - i]$$

$$= (w - 3)^2 - i^2$$

$$= w^2 - 6w + 10$$

$$5w^3 - 33w^2 + 68w - 30 = (w^2 - 6w + 10)(5w - 3) = 0$$

The other roots are $3 + i$ and $\frac{3}{5}$.

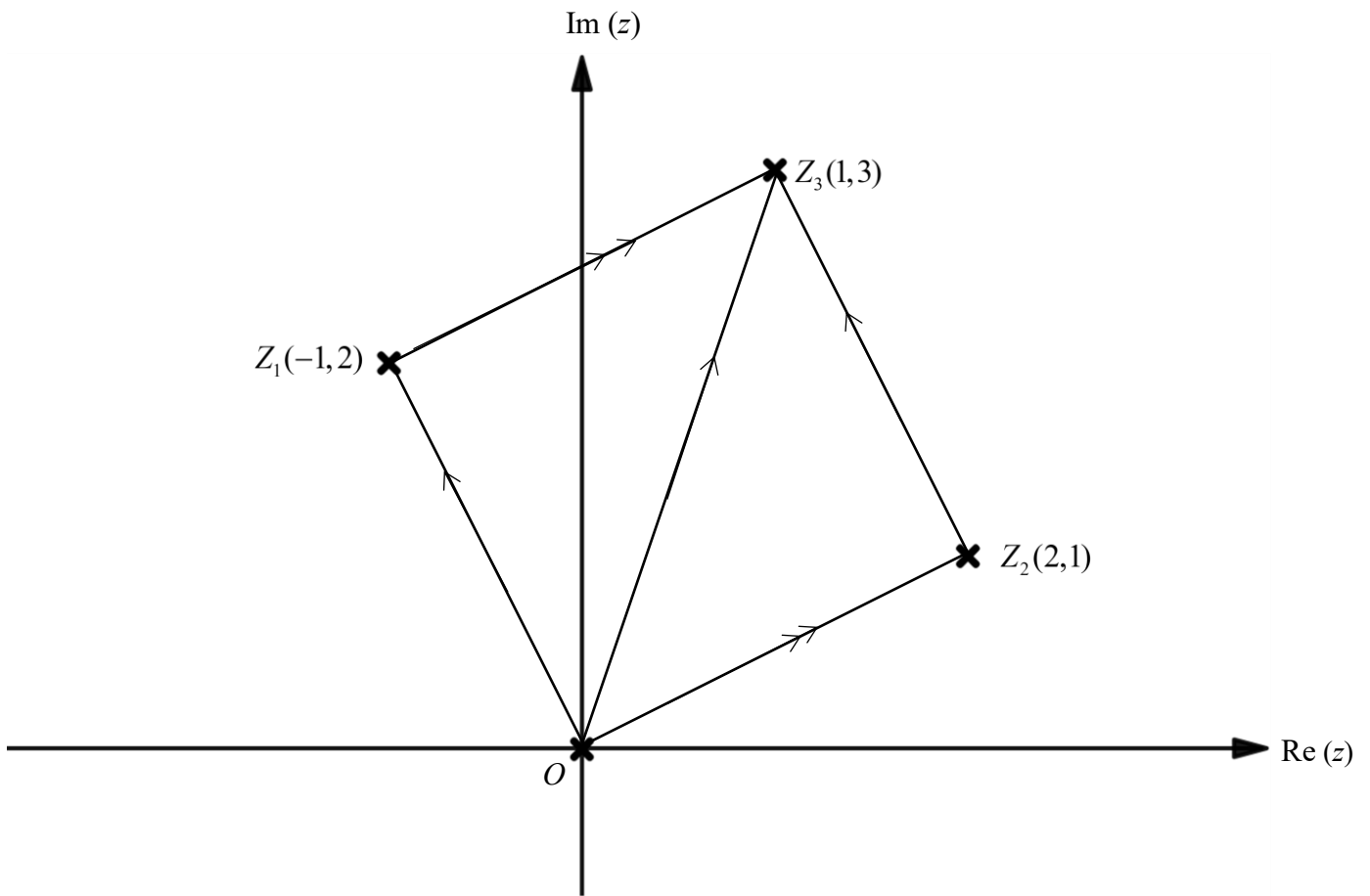
Use GC polyroot finder to check the answer.

NORMAL FLOAT AUTO a+bi RADIAN MP
 PLYSMT2 APP
 5x³- 33x²+ 68x- 30=0

 x1=3/5
 x2=3-1i
 x3=3+1i

[MAIN] [MODE] [COEFF] [STORE] [F] [D]

(b)



(i)

$$\frac{z_1}{z_2} = \frac{-1+2i}{2+i} \times \frac{2-i}{2-i} = \frac{-2+i+4i+2}{5} = i$$

(ii)

$$\frac{z_1}{z_2} = i \Rightarrow z_1 = iz_2$$

Rotate Z_2 by $\frac{\pi}{2}$ radians anti-clockwise about O to get Z_1

(1) $\therefore Z_1 O Z_2$ is a right angle.

(2) $OZ_1 = OZ_2$ (since $z_1 = iz_2$ or since $|z_1| = |-1+2i| = \sqrt{5}$ and $|z_2| = |2+i| = \sqrt{5}$)

(3) Also, since $\overrightarrow{OZ_3} = \overrightarrow{OZ_1} + \overrightarrow{OZ_2}$, $OZ_1 Z_3 Z_2$ is a parallelogram.

From (1) + (2) + (3), we can deduce that $OZ_2 Z_3 Z_1$ is a square.

Q5**(i)**

Number of ways to draw 4 orbs = ${}^{12}C_4 = 495$

(ii)

Number of ways to draw at least 2 colours = $495 - {}^5C_4 - {}^4C_4 = 489$

(iii)

Case 1: 2R, 1B, 1G

$$\text{Number of ways} = {}^5C_2 \times {}^4C_1 \times {}^3C_1 = 120$$

Case 2: 1R, 2B, 1G

$$\text{Number of ways} = {}^5C_1 \times {}^4C_2 \times {}^3C_1 = 90$$

Case 3: 1R, 1B, 2G

$$\text{Number of ways} = {}^5C_1 \times {}^4C_1 \times {}^3C_2 = 60$$

Total number of ways = $120 + 90 + 60 = 270$

Q6**(i)**

The team should conduct a 1-tail test as they are verifying the claim that users are spending more than 15 minutes.

(ii)

Central Limit Theorem states that sample means will follow a normal distribution approximately when the sample size is more than 30. Since the sample size is 80, the team is able to carry out a hypothesis test without knowing anything about the distribution of the times spend by the users.

(iii)

X : time spent per visit

μ : population mean time spent per visit

$$H_0 : \mu = 15$$

$$H_1 : \mu > 15$$

Under H_0 , since $n = 80$ is large, by Central Limit Theorem,

$$\bar{X} \sim N\left(15, \frac{\sigma^2}{80}\right) \text{ approximately}$$

$$\text{Test statistic, } z = \frac{\bar{x} - \mu}{\sqrt{\frac{\sigma^2}{n}}} = \frac{16 - 15}{\sqrt{\frac{\sigma^2}{80}}} = \frac{1}{\sqrt{\frac{\sigma^2}{80}}}$$

For 5% level of significance, reject H_0 if $z > 1.6449$

In order for platform's claim to be valid, H_0 must be rejected, hence,

$$\frac{1}{\sqrt{\frac{\sigma^2}{80}}} > 1.6449$$

$$\sqrt{\frac{\sigma^2}{80}} < 0.6079396$$

$$0 < \sigma < 5.44$$

(iv)

There is a probability of 0.05 of concluding that the population mean time spent per visit is more than 15 minutes when the population mean time spent is 15 minutes.

Q7

(i)

$$a = P(X = 0) = \left(\frac{1}{2} \times \frac{1}{2}\right) + \left(\frac{1}{4} \times \left(\frac{1}{2}\right)^2\right) + \left(\frac{1}{4} \times \left(\frac{1}{2}\right)^3\right) = \frac{11}{32}$$

$$b = P(X = 2) = \left(\frac{1}{4} \times \left(\frac{1}{2}\right)^2\right) + \left(\frac{1}{4} \times {}^3C_2 \times \left(\frac{1}{2}\right)^3\right) = \frac{5}{32}$$

Or

$$b = P(X = 2) = 1 - \frac{11}{32} - \frac{15}{32} - \frac{1}{32} = \frac{5}{32}$$

$$a = \frac{11}{32}, b = \frac{5}{32}$$

(ii)

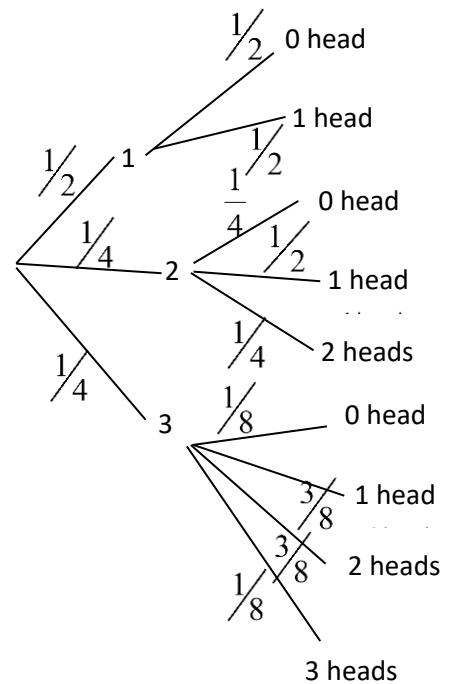
Probability distribution of X :

x	0	1	2	3
$P(X = x)$	$\frac{11}{32}$	$\frac{15}{32}$	$\frac{5}{32}$	$\frac{1}{32}$

$$E(X) = \sum xP(X = x) = (0)\left(\frac{11}{32}\right) + (1)\left(\frac{15}{32}\right) + (2)\left(\frac{5}{32}\right) + (3)\left(\frac{1}{32}\right) = \frac{7}{8}$$

$$E(X^2) = \sum x^2P(X = x) = (0)^2\left(\frac{11}{32}\right) + (1)^2\left(\frac{15}{32}\right) + (2)^2\left(\frac{5}{32}\right) + (3)^2\left(\frac{1}{32}\right) = \frac{11}{8}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{11}{8} - \left(\frac{7}{8}\right)^2 = \frac{39}{64} \text{ (Shown)}$$



(iii)

Let X = number of heads obtained in one game

$\bar{X}$ = mean number of heads obtained per game

Since $n = 50$ is large, by the Central Limit Theorem, $\bar{X} \sim N\left(\frac{7}{8}, \frac{39}{64}\right) = N\left(\frac{7}{8}, \frac{39}{3200}\right)$ approximately

$$P(\bar{X} > 1) = 0.12876 \approx 0.129$$

Q8

(i)

(1) The condition of a portable speaker is independent of the condition of any other portable speakers.

Or

Whether a portable speaker is faulty is independent of the condition of any other portable speakers.

(2) The probability that any portable speaker is faulty is constant.

(ii)

Let X = number of faulty portable speakers out of 24

$$X \sim B(24, 0.02)$$

$$P(X \leq 1) = 0.917387 \approx 0.917$$

(iii)

Method 1:

Let Y = number of substandard cartons out of 50

$$P(X > 1) = 1 - 0.917387 = 0.082613$$

$$Y \sim B(50, 0.082613)$$

$$P(Y = 0) = 0.013416 \approx 0.0134$$

Method 2:

$$\text{Required Probability} = [P(X \leq 1)]^{50} = [0.917387]^{50} = 0.0134$$

(iv)

Let F = the number of faulty flash drives out of 3

$$F \sim B(3, p)$$

$P(\text{batch is accepted})$

$$\begin{aligned}
 &= P(F=0) + P(F=1) \times P(F=0) \\
 &= (1-p)^3 + {}^3C_1 p^1 (1-p)^2 \times (1-p)^3 \\
 &= (1-p)^3 [1 + 3p(1-p)^2]
 \end{aligned}$$

At least 95% of the batches are accepted

$$(1-p)^3 [1 + 3p(1-p)^2] \geq 0.95$$

From the GC, $0 < p \leq 0.0703$

Q9

(i)

Given $P(A|B) = \frac{2}{5}$,

$$\frac{P(A \cap B)}{P(B)} = \frac{2}{5}$$

$$P(A \cap B) = \frac{2}{5} \times \frac{1}{3} = \frac{2}{15}$$

Also, $P(A \cup B) = \frac{3}{5}$

$$P(A) + P(B) - P(A \cap B) = \frac{3}{5}$$

$$P(A) + \frac{1}{3} - \frac{2}{15} = \frac{3}{5}$$

$$P(A) = \frac{2}{5}$$

Since $P(A) = P(A|B) = \frac{2}{5}$, events A and B are independent.

Or

Since $P(A) \cdot P(B) = \frac{2}{5} \times \frac{1}{3} = \frac{2}{15} = P(A \cap B)$, events A and B are independent.

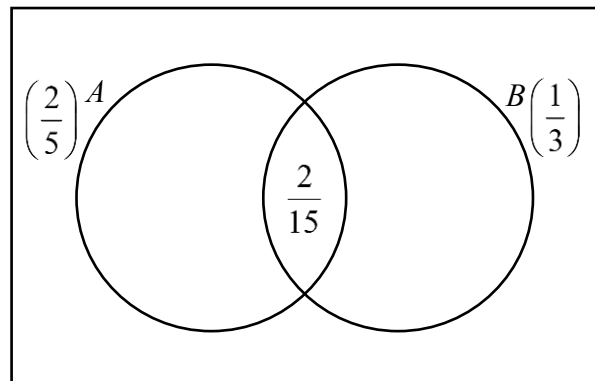
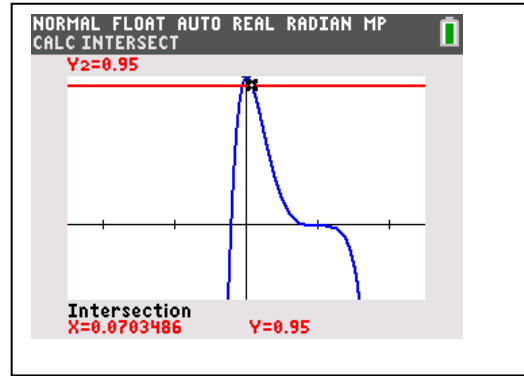
(ii)

$$P(A' \cap B) = P(A \cup B) - P(A) \quad \text{or} \quad P(A' \cap B) = P(B) - P(A \cap B)$$

$$= \frac{3}{5} - \frac{2}{5} = \frac{1}{5}$$

$$= \frac{1}{3} - \frac{2}{15} = \frac{1}{5}$$

Or since A and B are independent, A' and B are also independent.

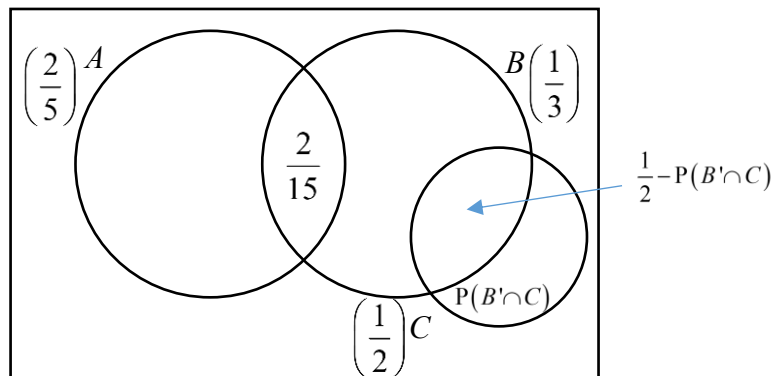


$$\begin{aligned}
 P(A' \cap B) &= P(A') \times P(B) \\
 &= \frac{3}{5} \times \frac{1}{3} = \frac{1}{5}
 \end{aligned}$$

(iii)

Method 1:

Since A and C are mutually exclusive, we have



From the Venn diagram,

$$P(B \cap C) \leq P(A' \cap B)$$

$$\frac{1}{2} - P(B' \cap C) \leq \frac{1}{5}$$

$$P(B' \cap C) \geq \frac{1}{2} - \frac{1}{5} = \frac{3}{10}$$

Also, $P(A \cup B \cup C) \leq 1$

$$P(A \cup B) + P(B' \cap C) \leq 1$$

$$\frac{3}{5} + P(B' \cap C) \leq 1$$

$$P(B' \cap C) \leq \frac{2}{5}$$

Hence, the least possible value of $P(B' \cap C) = \frac{3}{10}$

And the greatest possible value of $P(B' \cap C) = \frac{2}{5}$.

Method 2:

Let $P(B' \cap C) = x$

$$P(B \cap C) = \frac{1}{2} - x$$

$$P(A' \cap B \cap C') = P(A' \cap B) - P(B \cap C)$$

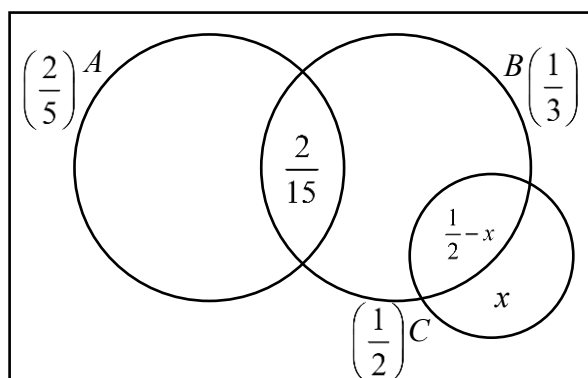
$$= \frac{1}{5} - \left(\frac{1}{2} - x\right)$$

$$= x - \frac{3}{10}$$

$$P(A' \cap B' \cap C') = 1 - P(A \cup B) - P(B' \cap C)$$

$$= 1 - \frac{3}{5} - x$$

$$= \frac{2}{5} - x$$



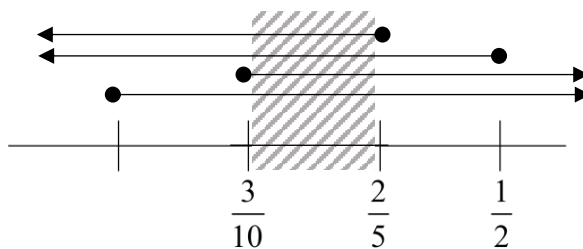
Since they are all probability, we have

$$x \geq 0$$

$$\frac{1}{2} - x \geq 0 \Rightarrow x \leq \frac{1}{2}$$

$$x - \frac{3}{10} \geq 0 \Rightarrow x \geq \frac{3}{10}$$

$$\frac{2}{5} - x \geq 0 \Rightarrow x \leq \frac{2}{5}$$



Solving the inequalities, $\frac{3}{10} \leq x \leq \frac{2}{5}$.

Hence, the least possible value of $P(B' \cap C) = \frac{3}{10}$

And the greatest possible value of $P(B' \cap C) = \frac{2}{5}$.

Q10

(i)

Let A = length of a ball of 2-ply yarn from Machine A

$$A \sim N(\mu, \sigma^2)$$

$$P(A < 340) = 0.8$$

$$P(A < 280) = 0.1$$

$$P\left(Z < \frac{340 - \mu}{\sigma}\right) = 0.8$$

$$P\left(Z < \frac{280 - \mu}{\sigma}\right) = 0.1$$

$$\frac{340 - \mu}{\sigma} = 0.84162$$

&

$$\frac{280 - \mu}{\sigma} = -1.28155$$

$$340 - \mu = 0.84162\sigma$$

$$280 - \mu = -1.28155\sigma$$

$$\mu + 0.84162\sigma = 340 \text{ -----(1)}$$

$$\mu - 1.28155\sigma = 280 \text{ -----(2)}$$

Solve (1) and (2)

$$\mu = 316.216 \approx 316$$

$$\sigma = 28.259 \approx 28.3 \quad (\text{AG})$$

(ii)

Let B = length of a ball of 3 ply yarn from Machine B

$$B \sim N(220, 6^2)$$

$$\begin{aligned} [P(B > 230)]^2 \times [P(B < 215)]^3 \times \frac{5!}{3!2!} &= 0.000189169 \\ &\approx 0.000189 \end{aligned}$$

(iii)

$$A \sim N(300, 20^2), \quad B \sim N(220, 6^2)$$

$$T = A_1 + A_2 + B_1 + B_2 + B_3 \sim N(2 \times 300 + 3 \times 220, 2 \times 20^2 + 3 \times 6^2) = N(1260, 908)$$

$$P(T > k) = 0.385$$

$$k = 1268.81 \quad (2\text{dp})$$

(iv)

$$A \sim N(300, 20^2), \quad B \sim N(220, 6^2)$$

$$X = 0.91(A_1 + A_2 + A_3) \sim N(0.91 \times 3 \times 300, 0.91^2 \times 3 \times 20^2) = N(819, 993.72)$$

$$Y = 0.95(B_1 + \dots + B_4) \sim N(0.95 \times 4 \times 220, 0.95^2 \times 4 \times 6^2) = N(836, 129.96)$$

$$X - Y \sim N(819 - 836, 993.72 + 129.96) = N(-17, 1123.68)$$

$$P(|X - Y| < 50) = P(-50 < X - Y < 50)$$

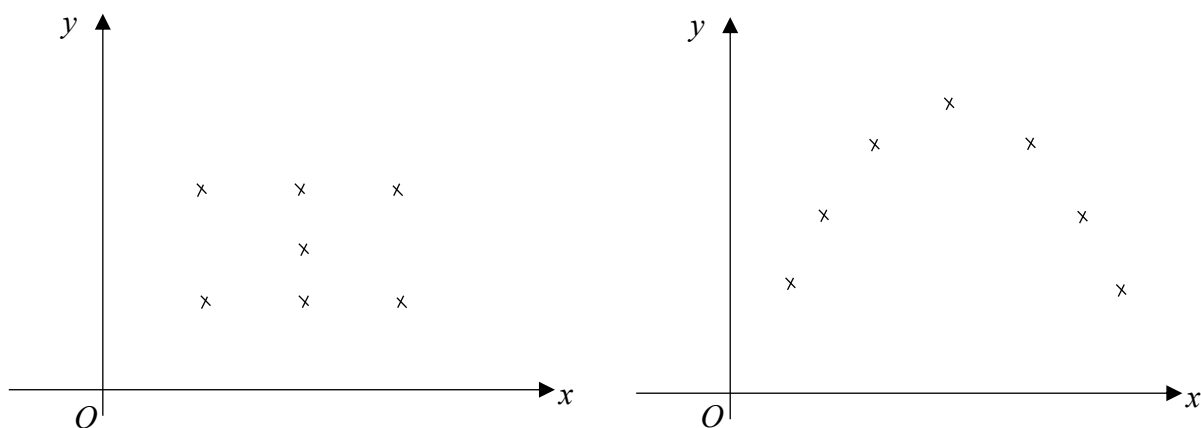
$$= 0.814733$$

$$\approx 0.815$$

11

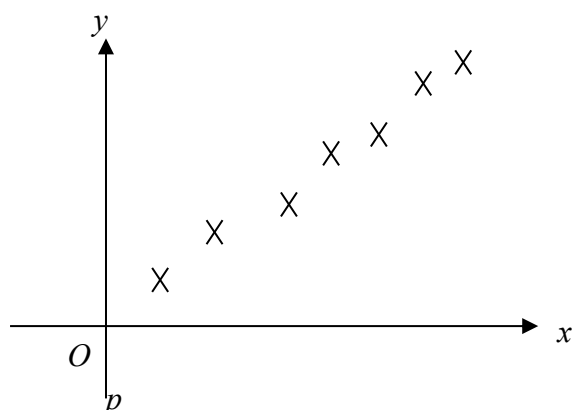
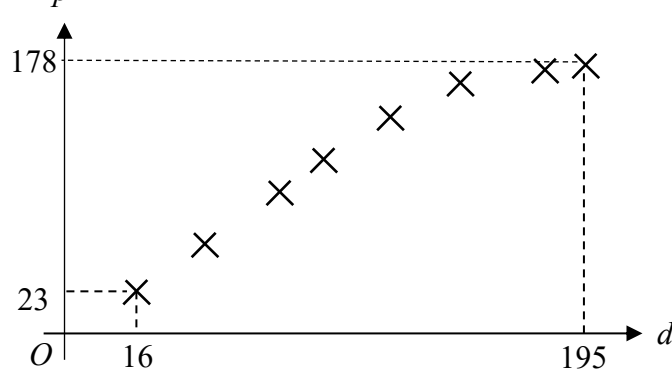
(a)(i)

Possible scatter diagrams:



(a)(ii)

One possible scatter diagram:

**(b)(i)****(b)(ii)**

The values of the residuals may be positive or negative. To avoid the cancellation effect in adding them, the squares of the residuals, which are non-negative, should be used.

(b)(iii)

The linear relationship is not appropriate as the scatter diagram indicates that as d increases, p increases at a decreasing rate.

(b)(iv)

$$p = -184.5953136 + 68.95820682 \ln d$$

$$\approx -185 + 69.0 \ln d \quad (\text{to 3 sf})$$

$$p = -184.5953136 + 68.95820682 \ln(210) \approx 184.13 \quad (\text{to 2 dp})$$

(b)(v)

Since the range of values of d is from 16 to 195, $d = 210$ is out of the given data range. To estimate the value of p when $d = 210$ is an extrapolation, which may not necessarily be reliable as the linear relationship between $\ln d$ and p may not hold at the extrapolated values.

(b)(vi)

The value of t will remain the same while the value of s will increase by 10.